

BUBT Intra University Programming Contest–BIUPC Summer 2026 (Junior)

<https://toph.co/c/bubt-intra-university-biupc-summer-2026-junior>



Schedule

The contest will run for **5h0m0s**.

Authors

The authors of this contest are Abid.699478, Arif_Khan, Djhoker, Goribulla, IFTEKHAR_AHAMED, Masud.0143, mdtuhinhasnat, Mubin_BUBT38, Polash_Islam, sahenulislam.274785, Soad21, Sohel_Rana, and user.089022.

Rules

This contest is formatted as per the official rules of ICPC Regional Programming Contests.

You can use C++17 GCC 13.2, C++20 Clang 16.0, C++20 GCC 13.2, C++23 GCC 13.2, C17 GCC 13.2, C23 GCC 13.2, Java 1.8, Kotlin 1.9, Kotlin 2.0, PyPy 7.3 (3.10), and Python 3.12 in this contest.

Be fair, be honest. Plagiarism will result in disqualification. Judges' decisions will be final.

Notes

There are 12 challenges in this contest.

Please make sure this booklet contains all of the pages.

If you find any discrepancies between the printed copy and the problem statements in Toph Arena, please rely on the latter.

A. Welcome to BUBT CP Community

Limits 1s, 256 MB

Some people think competitive programming is only for geniuses. The truth? It is for anyone who refuses to give up. Every senior who solves problems today once stared at their first problem, completely lost. The only secret is that they started and they kept going.

In the BUBT CP Community, we believe in one thing: every journey starts with a single Accepted. Today is that day for you. There are no tricky inputs, no hidden traps, and nothing to fear. There is only one simple line waiting to be printed.

So take a deep breath, open your editor, and print the message below. That is all it takes to take your first step with us.

Input

There is no input for this problem.

Output

Print a single line containing exactly:

Welcome to BUBT CP Community!

Example

Input	Output
No input is needed.	Welcome to BUBT CP Community!

B. Journey to Home

Limits 1s, 256 MB

Soad is finally going back home after a long time. On his way, he reaches a bridge where a gatekeeper stops him and asks for a 200 Taka toll. Soad checks his wallet and sees that he has N Taka. If Soad has enough money, he successfully reaches his home. If not, his journey gets blocked, and he has to return to his hostel.

Input

A single integer N ($0 \leq N \leq 10000$) the amount of money in Taka Soad has in his wallet.

Output

Print **Yes** if Soad can go to his home or **No** otherwise.

Examples

Input	Output
200	Yes

Input	Output
150	No

C. Hamba!

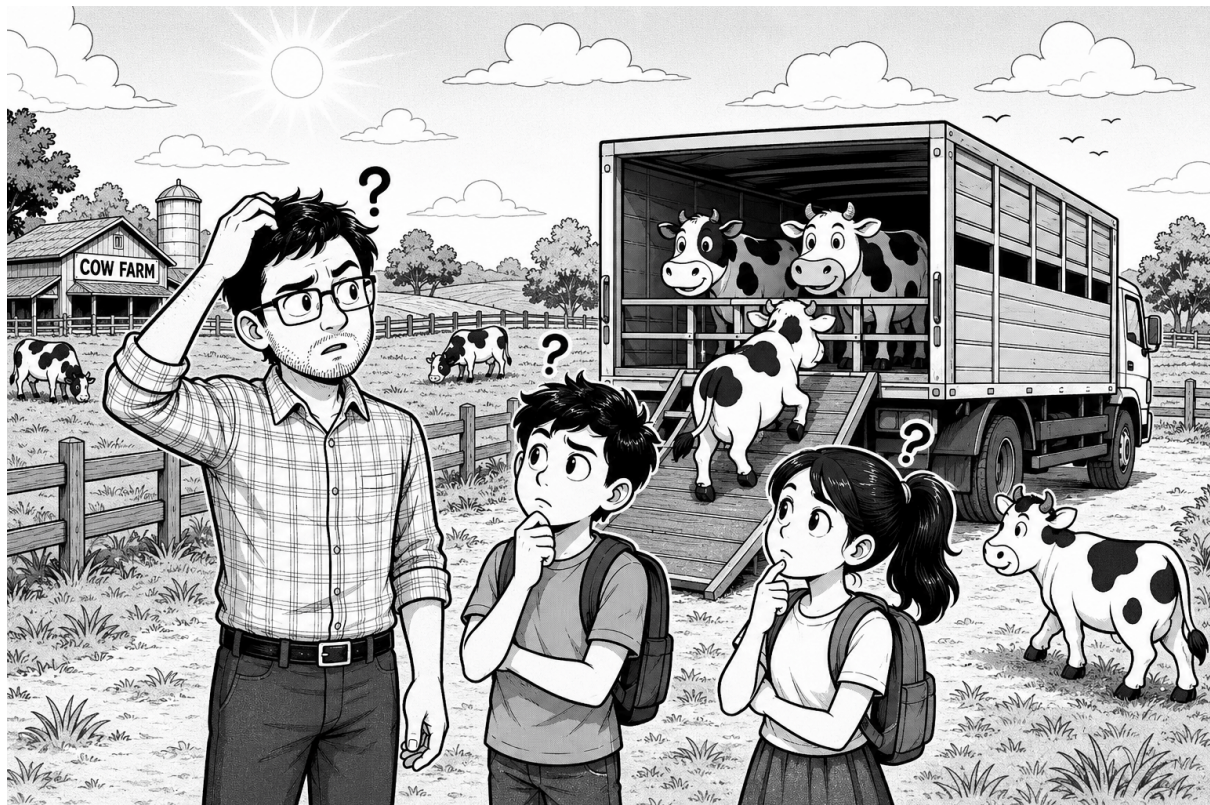
Limits 500ms, 256 MB

The BUBT CP Community is hosting a special contest! Forget about prize money or trophie this time, the winners are getting cows so they can start their very own cow farm!

The responsibility of managing this unique prize has been given to the famous cow businessman, Tasir Bhai, along with his business partners, Tasu and Miru.

There are N cows that need to be transported to the contest venue. To do this, they must rent trucks from the market. Each truck can carry a maximum of K cows, and renting a single truck costs C taka.

Tasu and Miru sat down to do the math, but they couldn't figure it out. Now, it's your turn to help Tasir Bhai's team! Can you find the minimum number of trucks needed and the total cost?



Input

The first and only line of input contains three space-separated integers N , K , and C ($1 \leq N, K, C \leq 10^9$), representing the total number of cows, the maximum number of cows a single truck can carry, and the rental cost of one truck, respectively.

Output

Print two space-separated integers: the minimum number of trucks required and the total rental cost in taka.

Example

Input	Output
10 5 10	2 20

D. Arif's Gold

Limits 1s, 256 MB

Mr. Arif received a grams of 24-carat gold from BIUPC, which became his only source of wealth. As a devout Muslim, he evaluates his wealth every year to pay Zakat. The Nisab (minimum threshold) for 24-carat gold is 85 grams. If Mr. Arif has at least 85 grams of gold at the time of his annual evaluation, he gives exactly 2.5% of his total current gold as Zakat. If his gold falls below 85 grams, he no longer pays Zakat. He lived for exactly x years after receiving the gold, meaning he performed this evaluation exactly x times.

After his passing, his children started a dispute over the distribution of the inheritance. To find a fair resolution, they came to the BUBT CP Community, knowing its members are wise enough to solve this.

According to Islamic inheritance laws, once a person's initial obligations are settled after their death, the remaining wealth is distributed among their children. If the person leaves behind at least one son and one daughter, all sons receive equal shares, and all daughters receive equal shares. The ratio of each son's share to each daughter's share is exactly 2 : 1, meaning that each son receives twice as much as each daughter.

Mr. Arif left behind b sons and g daughters. However, there is a technical issue: there is no equipment to measure the current weight of the gold in the BUBT CP lab. Your task is to mathematically calculate the remaining amount of gold Mr. Arif left behind after x years of potential Zakat deductions, and then determine the exact amount of gold each son and each daughter will receive.

Input

The input contains four integers a , x , b , and g ($1 \leq a \leq 10^9$, $1 \leq x \leq 10^9$, $1 \leq b, g \leq 100$) – the initial amount of gold in grams, the number of years Mr. Arif lived, the number of sons, and the number of daughters, respectively.

Output

Print two space-separated real numbers on a new line: the amount of gold a single son will receive, and the amount of gold a single daughter will receive. Your answer will be considered correct if its absolute or relative error does not exceed 10^{-6} .

Example

Input	Output
100 1 3 2	24.3750000000 12.1875000000

Notes

Here, Mr. Arif initially has 100 grams of gold. Since $100 \geq 85$, Zakat is applicable. After one year, 2.5% of the gold is deducted, leaving 97.5 grams of gold. There are 3 sons and 2 daughters. Since each son receives twice the share of each daughter, there are $3 \times 2 + 2 = 8$ total shares. Thus, each son receives 24.375 grams and each daughter receives 12.1875 grams.

E. Rupa and the Apple Orchard

Limits 1.5s, 256 MB

Rupa is a very hardworking and sweet girl. She has a huge apple orchard on the slope of a hill at one edge of the village. This time, there has been a bumper crop in the orchard! After picking the apples, Rupa has arranged all the apples in a row on a long conveyor belt. The total number of apples is n . But there is a small problem. Due to the weather, while some apples are completely fresh and juicy, some apples have rotted from the inside. Rupa checked each apple and wrote a 'quality value' on its body:

- If the apple is good and tasty, a positive value (for example: 5, 12, 100) is written on it.
- And if the apple is rotten or spoiled, a negative value (for example: -3, -10, -1) is written on it.

Today, a big buyer from the city has come to buy apples from Rupa's orchard. But he will not buy all the apples on the belt together. He will choose a specific range of the conveyor belt, for example, starting from the l -th apple to the r -th apple. But before buying, he is asking Rupa a question, "Well Rupa, tell me exactly how many good (positive) apples are there inside this l to r range?" The buyer started asking such a weird question repeatedly (a total of q times). Rupa is a single person; it is impossible for her to repeatedly count and calculate on such a huge belt. So she has rushed to you. Can you help Rupa with your programming skills?

Input

The first line contains a single integer t ($1 \leq t \leq 10^4$), the number of test cases.

The first line of each test case contains two integers n and q ($1 \leq n, q \leq 1.5 \times 10^6$), the total number of apples and the number of questions the buyer asked.

The second line of each test case contains n integers $v_1, v_2, \dots, v_n$ ($-10^9 \leq v_i \leq 10^9$, v_i not equal to 0) representing the quality value written on each apple.

Each of the next q lines of a test case contains two integers l and r ($1 \leq l \leq r \leq n$), the specific range chosen by the buyer.

It is guaranteed that the sum of n and the sum of q over all test cases do not exceed 1.5×10^6 .

Output

For each query, print a single integer on a new line — the exact number of good (positive) apples inside the range $[l, r]$.

Example

Input	Output
2	4
7 4	2
10 -5 15 20 -2 -8 50	0
1 7	1
2 4	1
5 6	1
3 3	
3 2	
-1 10 -5	
1 3	
2 2	

Notes

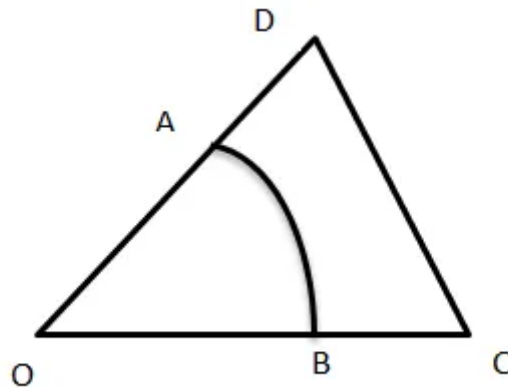
Explanation of the first testcase:

- Query 1 (1 7): Range from the 1st to the 7th apple. The good apples here are: 10, 15, 20, 50. A total of 4. So the answer is 4.
- Query 2 (2 4): 2nd to 4th apple. The values are: -5, 15, 20. There are 2 good apples here (15, 20). So the answer is 2.
- Query 3 (5 6): 5th to 6th apple. The values are: -2, -8. There are no good apples here. So the answer is 0.
- Query 4 (3 3): Only the 3rd apple. The value is 15, which is a good apple. So the answer is 1.

F. Geometry

Limits 1s, 512 MB

In geometry, an isosceles triangle is a triangle that has two sides of equal length. Sometimes it is specified as having exactly two sides of equal length, and sometimes as having at least two sides of equal length, the latter version thus including the equilateral triangle as a special case.



Centered scaled image.

You are given an isosceles triangle OCD where $OD = OC$ and $OA = OB$. As you are good in mathematics, you have to calculate the area of $ABCD$ where you will be given the angle of $\angle COD$ and the value of OD , and OA respectively.

Input

The first line of the input contains an integer T ($1 \leq T \leq 1000$), the number of test cases.

Each of the next T lines contains three floating-point numbers:

- $\angle COD$ (in radians), where $0 < \angle COD < \pi$;
- OA , where $0 \leq OA \leq 1000$;
- OC , where $OA \leq OC \leq 1000000$.

Output

For each test case, print Case X : followed by the area of $ABCD$, rounded to exactly 8 digits after the decimal point, where X is the test case number starting from 1.

Example

Input	Output
3 0.2 2 5 1 2 3 3 10 170	Case 1: 2.08336663 Case 2: 1.78661943 Case 3: 1889.18411647

Notes

The area of an isosceles triangle is $\frac{1}{2} \times ab \times \sin(\theta)$.

G. Odd and Prime Gold Rush

Limits 1s, 256 MB

In a small village in Bangladesh, there's a legendary treasure hidden in two ancient storerooms. One storeroom contains gold coins stacked in odd numbers, while the other storeroom holds coins stacked in prime numbers.

Every year during the village fair, the elders organize a challenge for adventurers. To unlock the main treasure chest, participants must solve a puzzle: they must calculate the absolute difference between the total coins in the first N stacks of both storerooms. Only the correct answer will reveal the treasure's secret.

Given N , find the absolute difference between:

- The sum of the first N odd numbers (from the Odd Storeroom).
- The sum of the first N prime numbers[†] (from the Prime Storeroom).

Will you be the one to solve the puzzle and claim the glory?

[†] A prime number is an integer strictly greater than 1 that is only divisible by 1 and itself. (e.g., 2, 3, 5, 7, 11).

Input

The only line of the input contains a single integer N ($1 \leq N \leq 10^3$) – the number of stacks to consider from each storeroom.

Output

Output a single integer – the absolute difference between the sum of the first N odd numbers and the sum of the first N prime numbers.

Examples

Input	Output
3	1

Input	Output
10	29

Notes

In the first sample test, the first 3 odd numbers are 1, 3, and 5. Their sum is $1 + 3 + 5 = 9$. The first 3 prime numbers are 2, 3, and 5. Their sum is $2 + 3 + 5 = 10$.

So, the absolute difference is $|9 - 10| = 1$.

H. The Lab Rats

Limits 1s, 256 MB

The BUBT Competitive Programming Community (CPC) wants to reward the top K students who spend the most time practicing in the competitive programming lab. Every time a student enters the lab, they tap their Student ID card on the RFID scanner.

You are given the log of ID card scans in exact chronological order (from oldest to newest). Generate a leaderboard based on these rules:

1. Rank students by their total number of lab visits (highest to lowest).
2. If there's a tie in visits, the student who visited most recently (closest to the end of the log) is ranked higher.

Print the Student IDs of the top K students in rank order. It is guaranteed there are at least K unique students in the log.

Input

The first line contains two integers N and K ($1 \leq K \leq N \leq 10^5$) – the total number of ID card scans in the log, and the number of top students to output.

The second line contains N space-separated integers $ID_1, ID_2, \dots, ID_N$ ($1 \leq ID_i \leq 10^5$), representing the Student IDs in the exact chronological order they were scanned.

Output

Print K space-separated integers on a single line – the Student IDs of the top K students in order of their rank.

Examples

Input	Output
7 3 101 205 101 303 205 101 404	101 205 404
Input	Output
5 2 55 55 55 99 88	55 88

Notes

For Sample 1: ID 101 has 3 visits, ID 205 has 2 visits, ID 404 has 1 visit (last seen at position 7), and ID 303 has 1 visit (last seen at position 4). The top 3 are 101, 205, and 404 because 404 visited more recently than 303.

For Sample 2: ID 55 has 3 visits, ID 88 has 1 visit (last seen at position 5), and ID 99 has 1 visit (last seen at position 4). The top 2 are 55 and 88 because 88 visited more recently than 99.

I. Family Tradition

Limits 1s, 256 MB

Mr. Kabiraj is participating in a national-level cooking competition, where he hopes to impress the judges by presenting his family's traditional recipe and ultimately win the championship. According to his mother, the secret to a perfect dish lies in balancing tastiness and beauty.

In the competition, he will be given n ingredients. For each ingredient i ($1 \leq i \leq n$):

- a_i represents its tastiness value.
- b_i represents the amount of beauty it contributes to the final dish.

Mr. Kabiraj may choose any non-empty subset of the ingredients. Suppose he selects k ingredients with indices $i_1, i_2, \dots, i_k$. The resulting dish has

$$\text{Tastiness} = a_{i_1} \& a_{i_2} \& \dots \& a_{i_k},$$

where "&" denotes the bitwise AND operation, and

$$\text{Beauty} = \sum_{j=1}^k b_{i_j}.$$

The judges compare dishes using the following rules:

1. A dish with greater tastiness is always considered better.
2. If multiple dishes have the same tastiness, the one with greater beauty is preferred.

Help Mr. Kabiraj determine the maximum possible beauty among all dishes having the maximum possible tastiness.

Here, $\&$ denotes the bitwise AND operator. The bitwise $\&$ operation compares two numbers bit by bit. In each bit position, the resulting bit is **1** if and only if both corresponding bits of the operands are **1**. Otherwise, the resulting bit is **0**.

Example: 5 (in binary: **101**) **3** (in binary: **011**) **5** $\&$ **3** = **1** (in binary: **001**)

Input

The first line contains a single integer t ($1 \leq t \leq 10^4$) – the number of test cases.

The description of each test case is as follows.

The first line contains a single integer n ($1 \leq n \leq 2 \times 10^5$) – the number of ingredients.

The second line contains n integers $a_1, a_2, \dots, a_n$ ($0 \leq a_i < 2^{30}$) – the tastiness values of the ingredients.

The third line contains n integers $b_1, b_2, \dots, b_n$ ($1 \leq b_i \leq 10^9$) – the beauty values of the ingredients.

It is guaranteed that the sum of n over all test cases does not exceed 2×10^5 .

Output

For each test case, print a single integer – the maximum possible beauty among all dishes having the maximum possible tastiness.

Example

Input	Output
3	3
2	4
4 1	7
3 4	
4	
1 2 3 4	
1 2 3 4	
5	
1 2 3 5 5	
2 4 3 5 2	

J. The Mad Scientist

Limits 1s, 256 MB

Dr. Blob is a mad scientist experimenting with n elements placed in a single row. The i -th element has a mass of a_i .

Dr. Blob wants to combine all n elements into a single, ultimate element. He can perform the following operation any number of times (possibly zero):

- Choose two adjacent elements, one with an odd mass x and the other with an even mass y , such that $x > y$.
- The two chosen elements are replaced by a single element with mass $x + y$.

Determine if Dr. Blob can achieve his goal of reducing all elements to one ultimate element.

Input

The first line contains an integer t ($1 \leq t \leq 10^4$) – the number of test cases.

The first line of each test case contains an integer n ($1 \leq n \leq 2 \cdot 10^5$) – the number of elements.

The second line contains n integers $a_1, a_2, \dots, a_n$ ($1 \leq a_i \leq 10^9$) – the masses of the elements.

It is guaranteed that the sum of n over all test cases does not exceed $2 \cdot 10^5$.

Output

For each test case, print "YES" (without quotes) if it is possible to reduce all the elements to exactly one, and "NO" (without quotes) otherwise.

You can output each letter in any case (for example, "YES", "Yes", "yes", "yEs" will all be recognized as a positive answer).

Example

Input	Output
2	YES
1	NO
42	
3	
10 5 2	

Notes

In the first test case, there is only one element. Dr. Blob is already done.

In the second test case, the array is $[10, 5, 2]$. The element with mass 5 merges with 2 to become 7. The array becomes $[10, 7]$. The element with mass 7 cannot merge with 10 because it is not strictly greater ($7 < 10$). No more operations can be performed, and the elements cannot be reduced to one.

K. Road to Master

Limits 2s, 512 MB

Rahad Maruf vai has been participating in online programming contests for years. His goal is to become a "Master" on a popular contest site. He realizes that, in order to achieve his target, he needs mastery in many different topics. So he decides to master the skill of XOR operations.

XOR is a bitwise operation that is performed on two bit patterns of equal length. Bit patterns are strings that contain only 0 or 1. The XOR operation produces another bit pattern of the same length. The i -th bit of the result is 1 if exactly one of the corresponding bits of the given patterns is 1; otherwise, it is 0.

For example, suppose we have two bit patterns $X = 101$ and $Y = 001$:

$$\begin{array}{r} 101 \\ \text{XOR } 001 \\ \hline 100 \end{array}$$

Any integer can be represented as a binary bit pattern. If necessary, leading zeros may be added so that two numbers have bit patterns of equal length.

After studying XOR, vai comes up with the following problem.

Given an array A of size N , count the number of pairs (i, j) ($1 \leq i \leq j \leq N$) such that the binary representation of $A_i \oplus A_j$ contains at least one bit equal to 1.

Help Maruf to solve this problem efficiently.

Best of Luck!

Input

The first line contains an integer N ($1 \leq N \leq 10^5$), the size of the array A

The next line contains N integers $A_1, A_2, \dots, A_N$ where $0 \leq A_i \leq 10^9$ for all $1 \leq i \leq N$

Output

Print a single integer — the number of pairs (i, j) ($1 \leq i \leq j \leq N$) such that $A_i \oplus A_j$ contains at least one bit equal to 1.

Examples

Input	Output
3 1 2 3	3

Input	Output
4 1 2 2 3	5

L. Flight Ledger

Limits 1s, 256 MB

US-Bangla Airlines recently announced the acquisition of 21 new Boeing aircraft in a deal worth approximately 1.5 billion. The new fleet will include 15 Boeing 737-8 aircraft and 6 Boeing 737-800 aircraft, all scheduled to be delivered in phases by the end of 2027 under a leasing programme.

As the airline prepares for this significant expansion, its operations team is reviewing passenger data from existing flights. For every flight operation, the team records its date, flight number, and passenger count.

The same flight number may appear on several different dates. For example, flight BS131 may carry passengers on multiple days. Management now needs reports for specific periods, such as the total number of passengers carried by BS131 during a particular week or month.

You are given all passenger records and Q requests. Each request specifies a flight number and a date range. For every request, determine the total number of passengers carried by that flight between the given dates, inclusive.

Input

The first line contains an integer N — the number of passenger records. Each of the next N lines contains *date*, *flightNumber*, and *paxCount*. The date is written in YYYY-MM-DD format. The next line contains an integer Q — the number of requests. Each of the next Q lines contains *flightNumber*, *startDate*, and *endDate*. Both dates are included in the requested range. It is guaranteed that $1 \leq N, Q \leq 2 \cdot 10^5$, $0 \leq paxCount \leq 10^9$, and $1 \leq |flightNumber| \leq 20$. Flight numbers contain only uppercase English letters and digits. All dates are valid, $startDate \leq endDate$, and every answer fits in a signed 64-bit integer.

Output

For each request, print one integer — the total number of passengers carried by the specified flight during the requested date range. Print the answers in the same order as the requests. If the flight has no passenger record in the requested range, print 0.

Example

Input	Output
10	505
2026-07-31 BS131 168	315
2026-07-28 BS205 150	357
2026-07-28 BS131 162	0
2026-08-02 BS131 190	0
2026-07-29 BS141 180	695
2026-07-29 BS131 175	
2026-07-30 BS205 155	
2026-08-01 BS205 160	
2026-07-30 BS141 172	
2026-08-03 BS141 185	
6	
BS131 2026-07-28 2026-07-31	
BS205 2026-07-29 2026-08-01	
BS141 2026-07-30 2026-08-03	
BS131 2026-08-01 2026-08-01	
BS999 2026-01-01 2026-12-31	
BS131 2026-01-01 2026-12-31	

Notes

In the first sample request, flight BS131 carried $162 + 175 + 168 = 505$ passengers from 2026-07-28 through 2026-07-31.

In the second sample request, only the records with 155 and 160 passengers for flight BS205 are inside the requested range, so the answer is 315.

There is no record for flight BS999, so its request produces 0.